11. If $f(x)=\dfrac{x}{x+1}$, then its differential is given by

(1) $\dfrac{-1}{(x+1)^2}dx$ (2) $\dfrac{1}{(x+1)^2}dx$ (3) $\dfrac{1}{x+1}dx$ (4) $\dfrac{-1}{x+1}dx$

12. If $u(x,y)=x^2+3xy+y-2019$, then $\left.\dfrac{\partial u}{\partial x}\right|_{(4,-5)}$ is equal to

(1) -4 (2) -3 (3) -7 (4) 13

13. Linear approximation for $g(x)=\cos x$ at $x=\dfrac{\pi}{2}$ is

(1) $x+\dfrac{\pi}{2}$ (2) $-x+\dfrac{\pi}{2}$ (3) $x-\dfrac{\pi}{2}$ (4) $-x-\dfrac{\pi}{2}$

14. If $w(x,y,z)=x^2(y-z)+y^2(z-x)+z^2(x-y)$, then $\dfrac{\partial w}{\partial x}+\dfrac{\partial w}{\partial y}+\dfrac{\partial w}{\partial z}$ is

(1) $xy+yz+zx$ (2) $x(y+z)$ (3) $y(z+x)$ (4) 0

15. If $f(x,y,z)=xy+yz+zx$, then f_x-f_z is equal to

(1) $z-x$ (2) $y-z$ (3) $x-z$ (4) $y-x$

SUMMARY

- Let $f:(a,b)\to\mathbb{R}$ be a differentiable function and $x_0\in(a,b)$ then linear approximation L of f at x_0 is given by
  $L(x)=f(x_0)+f'(x_0)(x-x_0)\forall x\in(a,b)$
- Absolute error =Actual value – Approximate value
  Relative error $=\dfrac{\text{Absolute error}}{\text{Actual value}}$
  Percentage error $=$ Relative error $\times 100$
  (or)
  $\dfrac{\text{Absolute error}}{\text{Acutal value}}\times 100$
- Let $f:(a,b)\to\mathbb{R}$ be a differentiable function. For $x\in(a,b)$ and Δx the increment given to x, the differential of f is defined by $df=f'(x)\Delta x$.
- All the rules for limits (limit theorems) for functions of one variable also hold true for functions of several variables.
- Let $A=\{(x,y)\,|\,a<x<b,\ c<y<d\}\subset\mathbb{R}^2, F:A\to\mathbb{R}$ and $(x_0,y_0)\in A$.

12th_Maths_Vol 2_EM_CH 8_Differentials and Partial.indd 88 16-12-2022 16:34:08